

MARKETING ANALYTICS

Data Analysis Report

SQL Queries · Python Sentiment Analysis · Power BI Dashboards

|

1. Executive Summary

This report presents a complete analytical pipeline for an online retail business experiencing declining customer engagement and conversion rates despite significant marketing investment. The analysis spans structured SQL data preparation, Python-based sentiment analysis on customer reviews, and interactive Power BI dashboards, providing evidence-based insights and actionable recommendations across four key areas: conversion funnel optimisation, content engagement, customer satisfaction, and geographic targeting.

2. Problem Statement

The business is facing three interconnected challenges undermining the return on its marketing investment:

- Reduced customer engagement — declining interactions with the site and marketing content.
- Decreased conversion rates — fewer site visitors are completing purchases.
- High marketing expenditure with below-expected returns — campaigns are not converting at anticipated rates.

The core objective is to identify which stages of the customer journey are failing, which content types drive engagement, and what customers are expressing through their reviews.

3. Key Performance Metrics

The following KPIs drive the entire analysis:

Metric	Definition	Business Purpose
Conversion Rate	% of visitors who make a purchase	Measures funnel effectiveness
Customer Engagement Rate	Clicks, likes & comments on content	Measures content relevance
Average Order Value (AOV)	Average spend per transaction	Measures revenue quality
Customer Feedback Score	Average rating from customer reviews	Measures satisfaction level

4. Analytical Goals & Insights

4.1 Increase Conversion Rates

Goal: Identify factors impacting conversion and recommend improvements to the conversion funnel.

- Pinpoint stages where visitors drop off (Awareness → Consideration → Purchase).
- Recommend targeted interventions such as retargeting campaigns or checkout optimisation.

4.2 Enhance Customer Engagement

Goal: Determine which content types generate the highest engagement across campaigns.

- Analyse views, clicks, and likes by content type (Social Media, Blog, Video).
- Inform future content strategy by identifying top-performing formats.

4.3 Improve Customer Feedback Scores

Goal: Understand recurring themes in customer reviews and surface actionable insights.

- Identify frequently mentioned product or service issues.
- Guide product improvements and customer service responses.

5. SQL Data Preparation

The following SQL queries clean, transform, and structure raw data from the MarketingAnalytics database into analysis-ready tables. Each section shows the query followed by its actual output.

5.1 Product Dimension (dim_product)

Products are categorised into Low / Medium / High price tiers to enable segmentation analysis:

Dim_product.sql

```
select * from dbo.products
--Query to categorise products based on their price

SELECT
    ProductID,
    ProductName,
    Price,
    CASE
        WHEN Price < 50 THEN 'Low'
        WHEN Price BETWEEN 50 AND 200 THEN 'Medium'
        ELSE 'High'
    END AS Price_Category
FROM dbo.products;
```

Query Output:

Results		Messages		
	ProductID	ProductName	Price	Price_Category
1	1	Running Shoes	223.75	High
2	2	Fitness Tracker	196.68	Medium
3	3	Yoga Mat	485.32	High
4	4	Dumbbells	26.21	Low
5	5	Soccer Ball	41.26	Low
6	6	Tennis Racket	36.07	Low
7	7	Basketball	225.12	High
8	8	Football Helmet	44.75	Low
9	9	Baseball Glove	327.36	High
10	10	Golf Clubs	81.59	Medium
11	11	Ski Boots	340.2	High
12	12	Ice Skates	37.56	Low
13	13	Swim Goggles	145.97	Medium
14	14	Cycling Helmet	472.32	High
15	15	Climbing Rope	410.17	High
16	16	Kayak	259.4	High
17	17	Surfboard	275.43	High
18	18	Volleyball	42.8	Low
19	19	Hockey Stick	173.83	Medium
20	20	Boxing Gloves	262.32	High

Figure 1 – Product dimension with Price_Category classification

5.2 Customer Dimension (dim_customers)

Customer records are enriched with geographic data via a LEFT JOIN on the geography table:

Dim_customers.sql

```
select * from dbo.customers
```

```
select * from dbo.geography
```

```
SELECT
    c.CustomerID,
    c.CustomerName,
    c.Email,
    c.Gender,
    c.Age,
    g.Country,
    g.City
FROM dbo.customers AS c
LEFT JOIN dbo.geography AS g
    ON c.GeographyID = g.GeographyID;
```

Query Output:

Results		Messages	
	GeographyID	Country	City
1	1	UK	London
2	2	Germany	Berlin
3	3	France	Paris
4	4	Spain	Madrid
5	5	Italy	Rome
6	6	Netherlands	Amsterdam
7	7	Belgium	Brussels
8	8	Sweden	Stockholm

	CustomerID	CustomerName	Email	Gender	Age	Country	City
1	1	Emma Anderson	emma.anderson@example.com	Male	50	Germany	Berlin
2	2	Sarah Brown	sarah.brown@example.com	Female	37	Spain	Madrid
3	3	Robert Hernan...	robert.hernandez@example.c...	Female	26	Netherl...	Amst...
4	4	David Garcia	david.garcia@example.com	Male	25	Sweden	Stoc...
5	5	Emma Miller	emma.miller@example.com	Female	41	Spain	Madrid
6	6	Daniel Rodrigu...	daniel.rodriguez@example.com	Male	31	Belgium	Brus...
7	7	Laura Miller	laura.miller@example.com	Male	35	Belgium	Brus...
8	8	James Gonzalez	james.gonzalez@example.com	Female	18	Sweden	Stoc...

Figure 2 – Customer dimension joined with geography data

Geographic concentration queries identify top markets:

```
--Write a SQL query to find the total number of customers in each country
--and display the country name along with the customer count in descending order.
```

```
SELECT
    g.Country,
    COUNT(c.CustomerID) AS Customers
FROM dbo.customers AS c
LEFT JOIN dbo.geography AS g
    ON c.GeographyID = g.GeographyID
GROUP BY g.Country
ORDER BY Customers DESC;
```

--Write a SQL query to list the top 5 cities with the highest number of customers, including the city name and total customers.

```
SELECT Top 5
    g.City,
    COUNT(c.CustomerID) AS Customers
FROM dbo.customers AS c
LEFT JOIN dbo.geography AS g
    ON c.GeographyID = g.GeographyID
GROUP BY g.City
ORDER BY Customers DESC;
```

Query Output:

	Country	Customers
1	Spain	18
2	Italy	12
3	Germ...	11
4	UK	10
5	Austria	10
6	Belgium	9
7	Nethe...	9
8	Swed...	8

	City	Customers
1	Madrid	18
2	Rome	12
3	Berlin	11
4	Lond...	10
5	Vienna	10

Figure 3 – Customers by country and top 5 cities by customer count

5.3 Customer Reviews Fact Table (fact_customer_reviews)

Review text is cleaned to remove double spaces before loading into the model:

```
select * from dbo.customer_reviews
```

```
Select
    ReviewID,
    CustomerID,
    ProductID,
    ReviewDate,
    Rating,
    Replace(ReviewText, ' ', ' ') as ReviewText
from dbo.customer_reviews;
```

Query Output:

Results		Messages				
	ReviewID	CustomerID	ProductID	ReviewDate	Rating	ReviewText
1	1	77	18	2023-12-23	3	Average experience, nothing special.
2	2	80	19	2024-12-25	5	The quality is top-notch.
3	3	50	13	2025-01-26	4	Five stars for the quick delivery.
4	4	78	15	2025-04-21	3	Good quality, but could be cheaper.
5	5	64	2	2023-07-16	3	Average experience, nothing special.
6	6	81	1	2025-12-21	4	Customer support was very helpful.
7	7	16	1	2024-01-29	3	Average experience, nothing special.
8	8	55	8	2024-08-15	5	The quality is top-notch.
9	9	3	13	2023-09-01	4	I love this product, will buy again!
10	10	78	6	2024-06-17	5	Excellent product, highly recommend!
11	11	54	9	2023-07-17	4	Great purchase, very satisfied.
12	12	77	2	2025-01-13	2	Product did not meet my expectations.
13	13	93	6	2024-08-20	5	Great purchase, very satisfied.
14	14	23	2	2025-06-16	4	Good quality, but could be cheaper.
15	15	29	11	2023-01-27	5	Great purchase, very satisfied.
16	16	95	17	2023-01-06	5	Excellent product, highly recommend!
17	17	88	4	2025-10-29	3	Not worth the money.
18	18	17	16	2024-11-23	4	Shipping was fast and the item was ...
19	19	34	13	2024-01-04	4	The quality is top-notch.
20	20	34	6	2023-04-29	1	I had a bad experience with this prod...
21	21	22	15	2023-04-08	5	Amazing value for the price.
22	22	52	3	2025-10-30	4	The quality is top-notch.
23	23	69	18	2023-07-02	5	Five stars for the quick delivery.

Figure 4 – Cleaned customer reviews fact table

5.4 Engagement Data Fact Table (fact_engagement_data)

Content type labels are standardised, the combined views/clicks field is split into two numeric columns, and Newsletter records are excluded:

```
select * from dbo.engagement_data

select
    EngagementID,
    ContentID,
    CampaignID,
    ProductID,
    UPPER(REPLACE(ContentType, 'Socialmedia', 'Social Media')) as ContentType,
    LEFT(ViewsClicksCombined, CHARINDEX('-', ViewsClicksCombined)-1) as Views,
    RIGHT(ViewsClicksCombined, Len(ViewsClicksCombined) - CHARINDEX('-', ViewsClicksCombined)) as Clicks,
    Likes,

    --Convert the Engagement Date to the dd.mm.yy format
    FORMAT(CONVERT(DATE, EngagementDate), 'dd.MM.yyyy') as EngagementDate

From dbo.engagement_data
where
    ContentType != 'Newsletter';
```

Query Output (before and after cleaning):

Results Messages									
	EngagementID	ContentID	ContentType	Likes	EngagementDate	CampaignID	ProductID	ViewsClicksCombined	
1	1	39	Blog	190	2023-08-30	1	9	1883-671	
2	2	48	Blog	114	2023-03-28	18	20	5280-532	
3	3	16	video	32	2023-12-08	7	14	1905-204	
4	4	43	Video	17	2025-01-21	19	20	2766-257	
5	5	16	newsletter	306	2024-02-21	6	15	5116-1524	
6	6	32	Socialmedia	648	2023-06-18	18	19	8237-1641	
7	7	33	SOCIALMEDIA	1	2025-10-01	12	2	750-34	
8	8	47	Blog	1	2025-03-31	17	6	891-35	
	EngagementID	ContentID	CampaignID	ProductID	ContentType	Views	Clicks	Likes	EngagementDate
1	1	39	1	9	BLOG	1883	671	190	30.08.2023
2	2	48	18	20	BLOG	5280	532	114	28.03.2023
3	3	16	7	14	VIDEO	1905	204	32	08.12.2023
4	4	43	19	20	VIDEO	2766	257	17	21.01.2025
5	6	32	18	19	SOCIAL M...	8237	1641	648	18.06.2023
6	7	33	12	2	SOCIAL M...	750	34	1	01.10.2025
7	8	47	17	6	BLOG	891	35	1	31.03.2025
8	9	48	13	16	BLOG	5571	1527	123	19.03.2024
9	10	4	15	15	BLOG	4279	297	25	03.12.2023
10	11	38	19	6	SOCIAL M...	4297	234	29	26.05.2024
11	12	29	3	10	SOCIAL M...	7877	2037	396	15.06.2023
12	13	14	12	18	VIDEO	1796	136	12	04.02.2025
13	14	2	16	2	SOCIAL M...	6837	2251	317	26.08.2023
14	15	16	6	19	SOCIAL M...	36	1	0	20.10.2025
15	16	2	17	16	SOCIAL M...	5980	1227	276	06.07.2023
16	17	35	18	10	BLOG	563	104	9	17.05.2024

Figure 5 – Engagement data: raw (top) and cleaned (bottom) with split Views/Clicks columns

5.5 Customer Journey Fact Table (fact_customer_journey)

Duplicate journey records are identified using ROW_NUMBER() and removed. Missing duration values are imputed with the daily average:

```
select * from dbo.customer_journey
```

```
With DuplicateRecords as (
    Select
        JourneyID,
        CustomerID,
        ProductID,
        VisitDate,
        Stage,
        Action,
        Duration,
        ROW_NUMBER() over(
            PARTITION by CustomerID,ProductID, VisitDate,Stage, Action
            ORDER by JourneyID
        ) As Row_numb
    From
        dbo.customer_journey
)
Select *
from DuplicateRecords
Where Row_numb>1
Order by JourneyID

--Remove Duplicates and Clean data

Select
    JourneyID,
    CustomerID,
    ProductID,
    VisitDate,
    Stage,
    Action,
    COALESCE( Duration, avg_duration) as Duration
From
    (
        Select
            JourneyID,
            CustomerID,
            ProductID,
            VisitDate,
            Upper(Stage) as STAGE,
            Action,
            Duration,
            avg(Duration) OVER (PARTITION BY VisitDate) as avg_duration,
            ROW_NUMBER() OVER (
                PARTITION BY CustomerID, ProductID, VisitDate, Upper(Stage), Action
                ORDER BY JourneyID
            ) as row_num
        From
            dbo.customer_journey
    ) As Sub_que
Where
    row_num = 1;
```

Query Output:

Results Messages							
	JourneyID	CustomerID	ProductID	VisitDate	Stage	Action	Duration
1	2040	1	1	2023-03-11	HOMEPAGE	Click	26
2	3077	1	1	2024-03-18	HOMEPAGE	View	66
3	3388	1	1	2024-06-28	HOMEPAGE	Click	18
4	577	1	1	2025-12-17	HOMEPAGE	Click	115
5	3312	1	2	2023-03-03	PRODUCT...	View	40
6	3696	1	2	2024-02-10	PRODUCT...	View	231
7	2124	1	3	2023-09-23	PRODUCT...	View	161
8	1548	1	3	2024-11-06	HOMEPAGE	View	238
9	2696	1	3	2025-07-09	PRODUCT...	View	110
10	3344	1	5	2023-07-20	CHECKOUT	Dro...	187.6
11	2490	1	6	2023-02-19	PRODUCT...	View	196
12	1929	1	6	2025-06-25	CHECKOUT	Dro...	185.4
13	55	1	7	2024-03-17	CHECKOUT	Dro...	226
14	2271	1	7	2025-09-17	CHECKOUT	Dro...	209
15	2187	1	9	2024-10-23	CHECKOUT	Purc...	10
16	1451	1	9	2025-01-25	CHECKOUT	Dro...	144
17	2473	1	10	2023-02-12	CHECKOUT	Purc...	271
18	1649	1	10	2023-06-21	CHECKOUT	Dro...	74.75
19	380	1	11	2023-05-29	CHECKOUT	Dro...	198
20	2265	1	11	2024-09-01	HOMEPAGE	View	15
21	3803	1	11	2025-08-10	PRODUCT...	View	283
22	685	1	12	2023-07-11	HOMEPAGE	View	22
23	1904	1	14	2025-11-15	HOMEPAGE	View	110
24	3174	1	15	2023-05-01	HOMEPAGE	View	239
25	2556	1	15	2024-06-15	PRODUCT...	View	125

Figure 6 – Deduplicated and cleaned customer journey fact table

6. Sentiment Analysis (Python / VADER)

Customer review sentiment is quantified using NLTK's VADER (Valence Aware Dictionary and sEntiment Reasoner), which is well-suited to short, informal product review text.

6.1 Methodology

- Reviews are fetched directly from SQL Server using pyodbc.
- VADER produces a compound score in the range [-1, +1]. Values above 0.05 indicate positive sentiment; below -0.05 indicate negative sentiment.
- A categorisation function cross-references the VADER score with the star rating to produce a nuanced sentiment label.
- A bucketing function groups scores into four ranges for aggregated dashboard reporting.

6.2 Sentiment Categories

Sentiment Label	VADER Score Condition	Star Rating
Positive	> 0.05	4 or 5 stars
Mixed Positive	> 0.05	3 stars
Neutral	-0.05 to 0.05	Any rating
Mixed Negative	< -0.05	3 stars
Negative	< -0.05	1 or 2 stars

6.3 Score Buckets

- 0.5 to 1.0 — Strongly positive sentiment
- 0.0 to 0.49 — Mildly positive sentiment
- -0.49 to 0.0 — Mildly negative sentiment
- -1.0 to -0.5 — Strongly negative sentiment

6.4 Python Output

The enriched dataset with SentimentScore, CategoryofSentiment, and SentimentBucket columns, exported as a CSV for Power BI ingestion:

ReviewID	CustomerID	ProductID	ReviewDate	Rating	ReviewText	SentimentScore	CategoryofSentiment	SentimentBucket
0	1	77	18 2023-12-23	3	Average experience, nothing special.	-0.3089	Mixed Negative	-0.49 to 0.0
1	2	80	19 2024-12-25	5	The quality is top-notch.	0.0000	Positive	0.0 to 0.49
2	3	50	13 2025-01-26	4	Five stars for the quick delivery.	0.0000	Positive	0.0 to 0.49
3	4	78	15 2025-04-21	3	Good quality, but could be cheaper.	0.2382	Mixed Positive	0.0 to 0.49
4	5	64	2 2023-07-16	3	Average experience, nothing special.	-0.3089	Mixed Negative	-0.49 to 0.0

```
cus_review_data.to_csv('fact_customer_reviewa_with_sentiment', index = False)
```

Figure 7 – Python sentiment analysis output with scored and categorised review records

7. Power BI Dashboards

The cleaned and enriched data is loaded into Power BI to produce four interactive dashboard pages. Each page is shown below as exported from Power BI.

7.1 Marketing Analytics Overview

The overview page presents high-level KPIs: 9.6% overall conversion rate, 9M social media views, 2M clicks, 414K likes, and a 3.69 average customer rating. Supporting visuals show conversion rate by product and by month, engagement trend lines, and average rating by product.

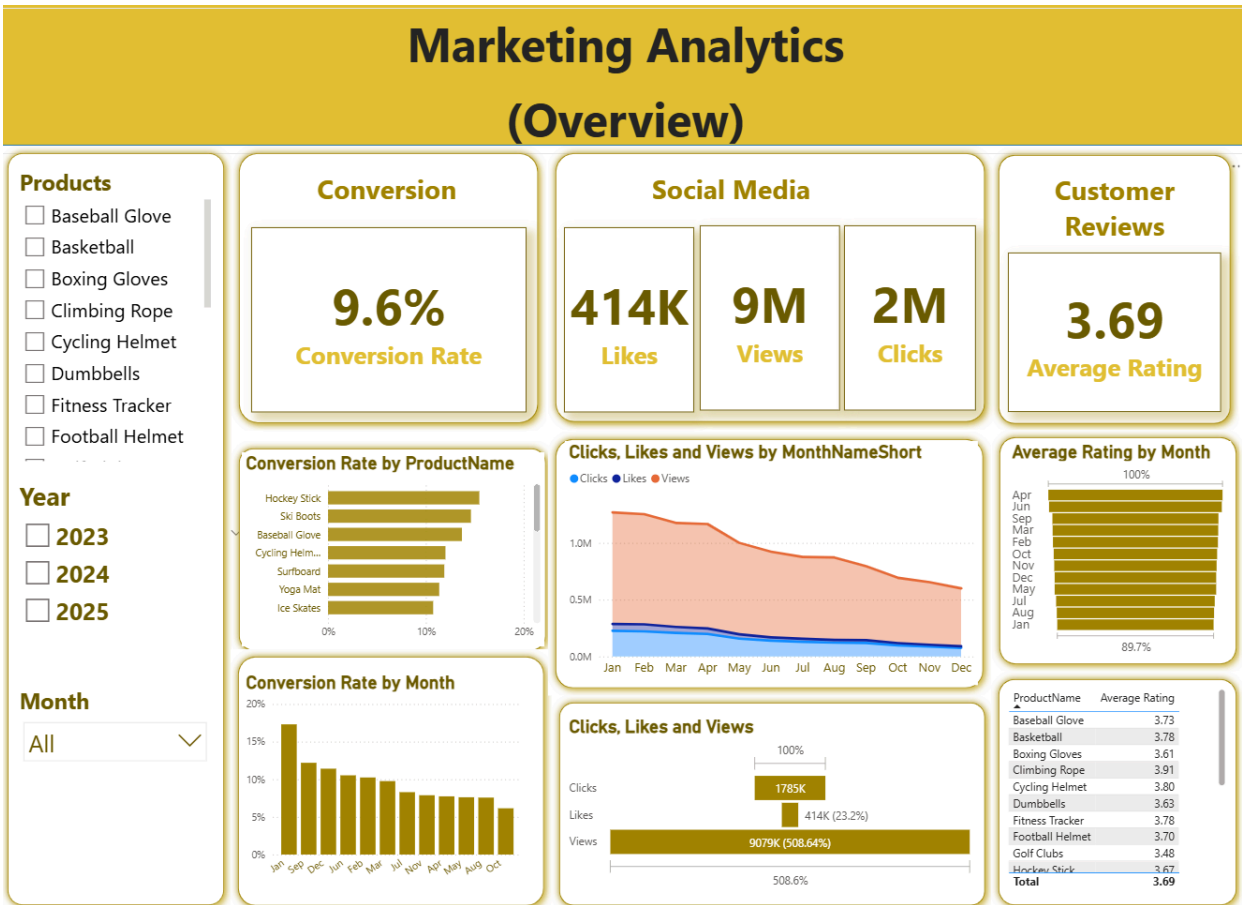
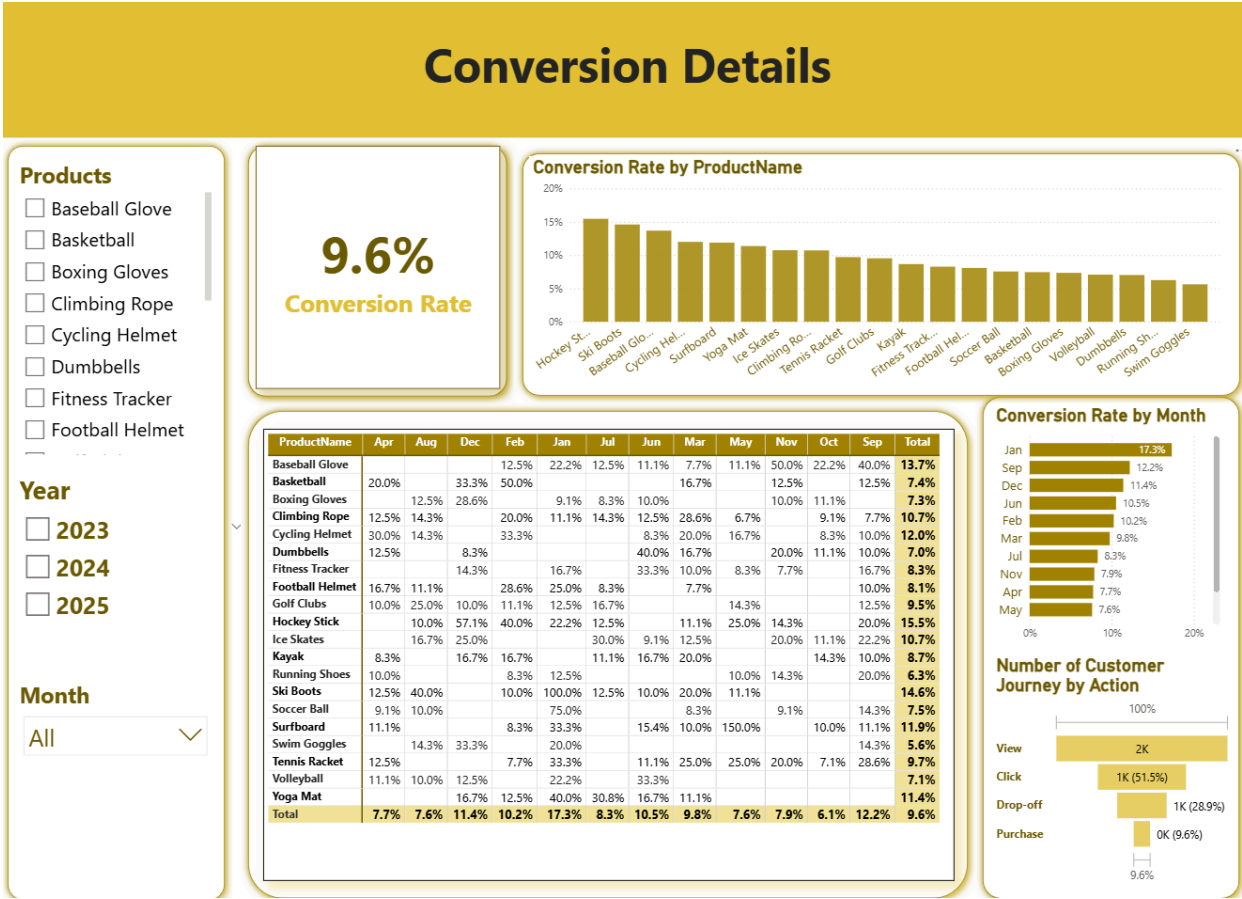
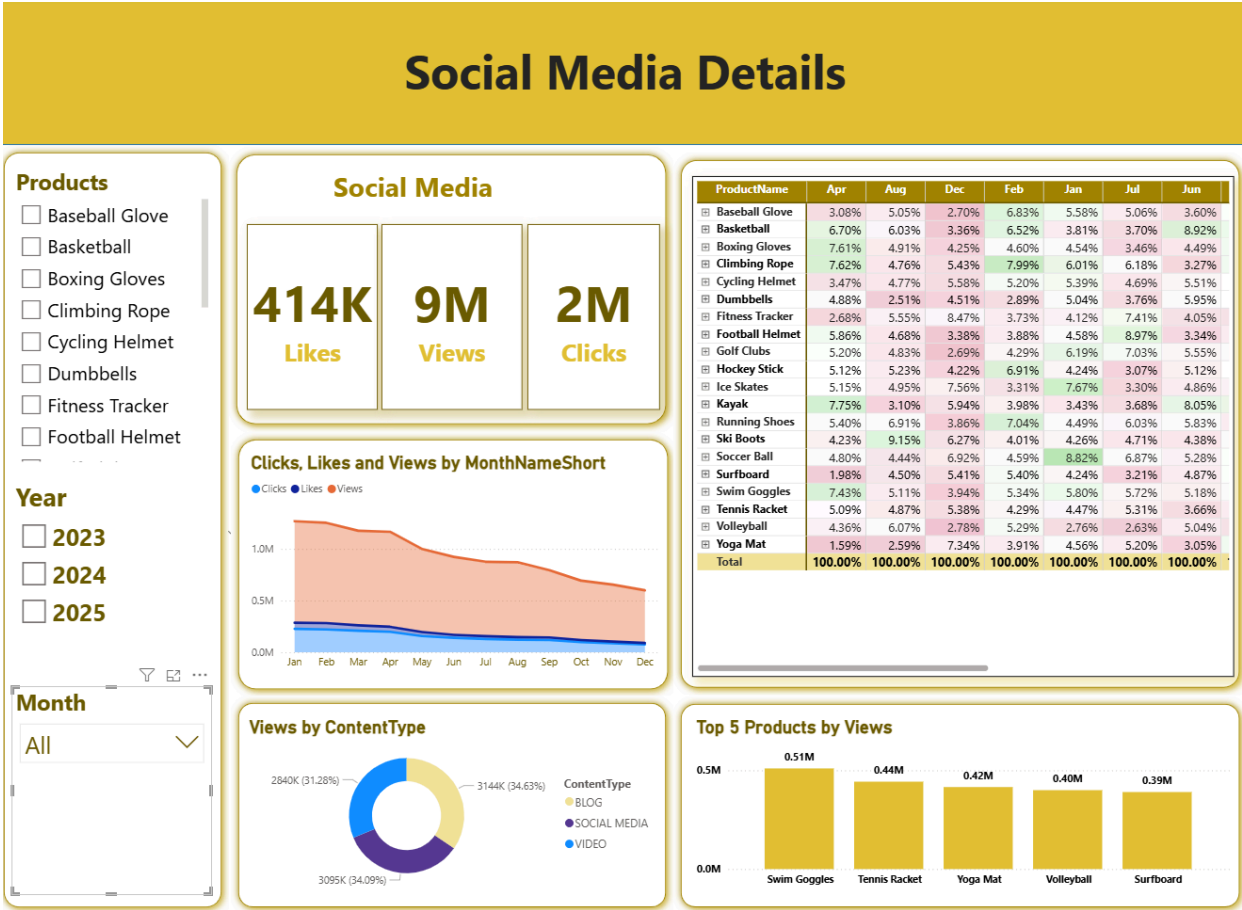


Figure 8 – Power BI: Marketing Analytics Overview dashboard

7.2 Conversion Details

This page deep-dives into the conversion funnel. A product-level matrix shows monthly conversion rates, the bar chart ranks products by overall conversion, and the journey action breakdown reveals that 51.5% of actions are Clicks, 28.9% are Drop-offs, and only 9.6% result in Purchases.





Customer Review Details

Sentiment Categories

- ☐ Mixed Negative
- ☐ Mixed Positive
- ☐ Negative
- ☐ Neutral
- ☐ Positive

Products

- ☐ Baseball Glove
- ☐ Basketball
- ☐ Boxing Gloves
- ☐ Climbing Rope
- ☐ Cycling Helmet
- ☐ Dumbbells

Year

All

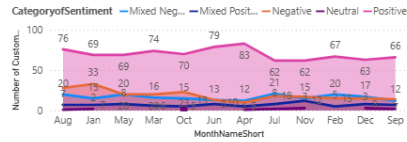
Month

All

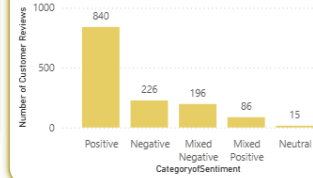
Customer Reviews

3.69
Average Rating

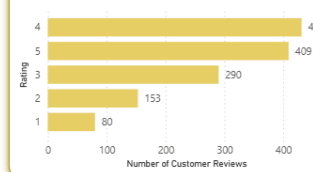
Monthly Customer Reviews by Sentiment



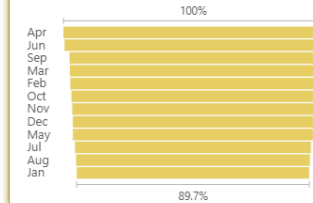
Number of Customer Reviews by Category of Sentiment



Number of Customer Reviews by Rating



Average Rating by Month



CustomerID	ReviewDate	ReviewText	CategoryofS
19	Wednesday, December 31, 2025	Five stars for the quick delivery.	Positive
30	Wednesday, December 31, 2025	Average experience, nothing special.	Mixed Nega
88	Tuesday, December 30, 2025	The quality is top-notch.	Positive
73	Sunday, December 28, 2025	Five stars for the quick delivery.	Positive
36	Saturday, December 27, 2025	Customer support was very helpful.	Positive
97	Friday, December 26, 2025	Five stars for the quick delivery.	Positive
75	Thursday, December 25, 2025	Excellent product, highly recommend!	Positive
68	Wednesday, December 24, 2025	Five stars for the quick delivery.	Positive
82	Wednesday, December 24, 2025	Exceeded my expectations!	Positive
97	Tuesday, December 23, 2025	Customer support was very helpful.	Positive
29	Monday, December 22, 2025	Customer support was very helpful.	Positive
30	Monday, December 22, 2025	The quality is top-notch.	Positive
43	Monday, December 22, 2025	Average experience, nothing special.	Mixed Nega
81	Sunday, December 21, 2025	Customer support was very helpful.	Positive
55	Saturday, December 20, 2025	I love this product, will buy again!	Positive
57	Saturday, December 20, 2025	Not worth the money.	Negative
60	Friday, December 19, 2025	Not worth the money.	Mixed Nega
72	Friday, December 19, 2025	Shipping was fast and the item was well-packaged.	Positive
38	Thursday, December 18, 2025	Exceeded my expectations!	Positive
100	Thursday, December 18, 2025	Not worth the money.	Mixed Nega
68	Tuesday, December 16, 2025	The build quality is impressive.	Positive
11	Monday, December 15, 2025	Exceeded my expectations!	Positive
32	Monday, December 15, 2025	Good quality, but could be cheaper.	Mixed Positi
42	Monday, December 15, 2025	Five stars for the quick delivery.	Positive
43	Sunday, December 14, 2025	Disappointed with the performance.	Negative
72	Saturday, December 13, 2025	Customer support was very helpful.	Mixed Positi

Figure 11 – Power BI: Customer Review Details dashboard

8. Recommendations

8.1 Conversion Funnel

- With 28.9% of journey actions being Drop-offs, prioritise the checkout and product page experience through A/B testing of CTAs, pricing, and page layout.
- Implement automated retargeting for visitors who reach the consideration stage but do not convert.
- Hockey Stick (15.5%) and Ski Boots (14.6%) have the highest conversion rates — analyse their product pages to replicate successful patterns across lower-converting products.

8.2 Content Strategy

- All three content types (Blog, Social Media, Video) have roughly equal reach (~34% share), so investment should be guided by cost-per-engagement rather than volume alone.
- Views have trended downward across all months — investigate whether campaign frequency, creative fatigue, or budget changes are the cause.
- Newsletters are excluded from engagement KPIs as they operate on a different distribution model and should be tracked separately.

8.3 Customer Satisfaction

- With 226 negative reviews (16.6% of total), address recurring themes through product improvements or clearer product descriptions.
- Proactively follow up with customers leaving mixed or negative reviews to recover satisfaction and prevent churn.
- The 3.69 average rating — pulled down by a concentration of 2-star and 1-star reviews — indicates room for improvement in fulfilment, product quality communication, and post-purchase support.

8.4 Geographic Targeting

- Spain (18 customers) and Italy (12) lead by country; Madrid (18) and Rome (12) lead by city — allocate higher budgets to these markets.
- Explore expansion into countries with growing but under-served customer bases to increase total addressable reach.

9. Conclusion

This report establishes a complete end-to-end analytical pipeline — from raw SQL data extraction and cleaning, through Python-powered VADER sentiment analysis, to four Power BI dashboard pages covering conversion, engagement, and customer reviews. The headline KPIs (9.6% conversion rate, 3.69 average rating) provide a baseline against which all future optimisation efforts can be measured. By acting on the recommendations across conversion funnel, content strategy, customer satisfaction, and geographic targeting, the business can make evidence-based improvements to its marketing strategy and measurably increase both engagement and conversion rates.